

NANAKJOT SINGH CHAHAL

+91-9988444520 | nanak006chahal@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

PROJECTS

Asendia — Real-time AI Voice Interviewer

Tech: Python, FastAPI, WebSockets, Gemini Live (speech-to-speech), Gemini Flash, Supabase, Web Audio API

- Full-duplex voice interviewer for ML/AI roles on Gemini Live: open mic, no STT/TTS hops, barge-in flushes queued audio instantly. Replaced a cascaded V3 (Groq Llama 3.3 70B + Whisper + ElevenLabs) that hit **~0.5–0.8s** replies on speculation hits.
- Persona encodes **6 ML bluff patterns** (concept mix-ups, metric confusion, impossible scale claims) and ends the interview itself via an end-interview tool call, backed by a **12-turn** hard cap and tool/scoring fallbacks so every session ends in a report.
- Conversation/judgment split: one post-call Gemini Flash JSON pass scores the transcript blind on **5 dimensions** (0–10) with an advance/reject/hold verdict; session resumption, exponential-backoff reconnects, and context compression keep long calls alive.

Stacy — AI Phone Receptionist

Tech: Python, LiveKit Agents, Twilio SIP, Deepgram Nova-3, Claude Haiku 4.5, ElevenLabs Flash v2.5, Silero VAD

- Inbound phone receptionist for US small businesses: Twilio SIP trunk straight into LiveKit (no WebSocket relay), streaming STT → LLM → TTS with barge-in, 6 intake/booking tools, and multi-tenant configs; **0.95s** p50 / **1.5s** p90 reply latency.
- Deterministic “**desk**” layer answers scripted turns (phone numbers, names, read-back confirmations) in Python before the LLM, in **~0.4s**; adaptive endpointing (0.20–0.45s VAD wait + semantic turn detector) prevents mid-address cut-offs.
- Traced three hidden orchestration stalls in framework source: default strict tool schemas (first token **4.3s** → **0.8s**), a markdown filter holding speech behind tool calls (up to **414ms** saved), and a **607ms** SSL import blocking the event loop (moved to prewarm).
- Benchmarks run from the us-east worker on true caller wait (stops speaking → agent speaks), one variable at a time; a full-call simulator gates model choice on behaviour, catching a 2×-faster model inventing a phone number.

Conflict Arbitration Environment

Tech: Multi-Agent RL, OpenEnv, HuggingFace TRL, Qwen 2.5, LoRA + GRPO, FastAPI

- OpenEnv-compliant RL environment where a referee agent watches two parallel AI agents and intervenes on spec drift.
- Designed a **4-phase curriculum** across **8 conflict types** and **8 domains** with programmatic contrastive rewards.
- Fine-tuned Qwen 2.5-1.5B via LoRA + GRPO on one A10G: custom training loop, reward/accuracy telemetry, 2,000-step run.

Reflux — MCP Reliability Tester

Tech: React, TypeScript, Vite, Tailwind, Express.js, Google Gemini

- Tests MCP tool definitions by running an AI agent against them; returns an **Agent Readiness Score (0–100)**.
- Four weighted rubrics — Runtime (40), Parameters (30), Docs (20), Stability (10) — pinpoint where a tool spec fails.
- Auto-generates optimized tool definitions and a fix report targeting naming, parameter ambiguity, and description gaps.

EDUCATION

Birla Institute of Technology and Science (BITS Pilani)

2024 – 2028

B.Sc. (Hons.) in Computer Science

Pilani, India

- CGPA: **8.82**

TECHNICAL SKILLS

Languages	Python, Java, JavaScript, TypeScript, SQL, HTML/CSS
AI & ML	LLM Integration, Prompt Engineering, RAG, Multi-Agent Systems, RL (GRPO, LoRA), HF TRL
Voice AI	Gemini Live (speech-to-speech), LiveKit Agents, Deepgram, ElevenLabs, Twilio SIP, barge-in handling
Backend & Frameworks	FastAPI, Node.js, Express.js, Spring Boot, React, Tailwind CSS, WebSockets, REST APIs
Databases & Data	PostgreSQL, MySQL, MongoDB, Supabase, Firebase, Pandas, Scikit-learn
Core Concepts	Data Structures & Algorithms, System Design (HLD), OOP, SEO / GEO
Tools	Git, Docker, Vercel, Postman, JWT Auth

ACHIEVEMENTS

- **Meta × PyTorch OpenEnv Hackathon (Apr 2026)**: Led team to the Top 100 of 31,000+ teams, 48-hour Grand Finale.